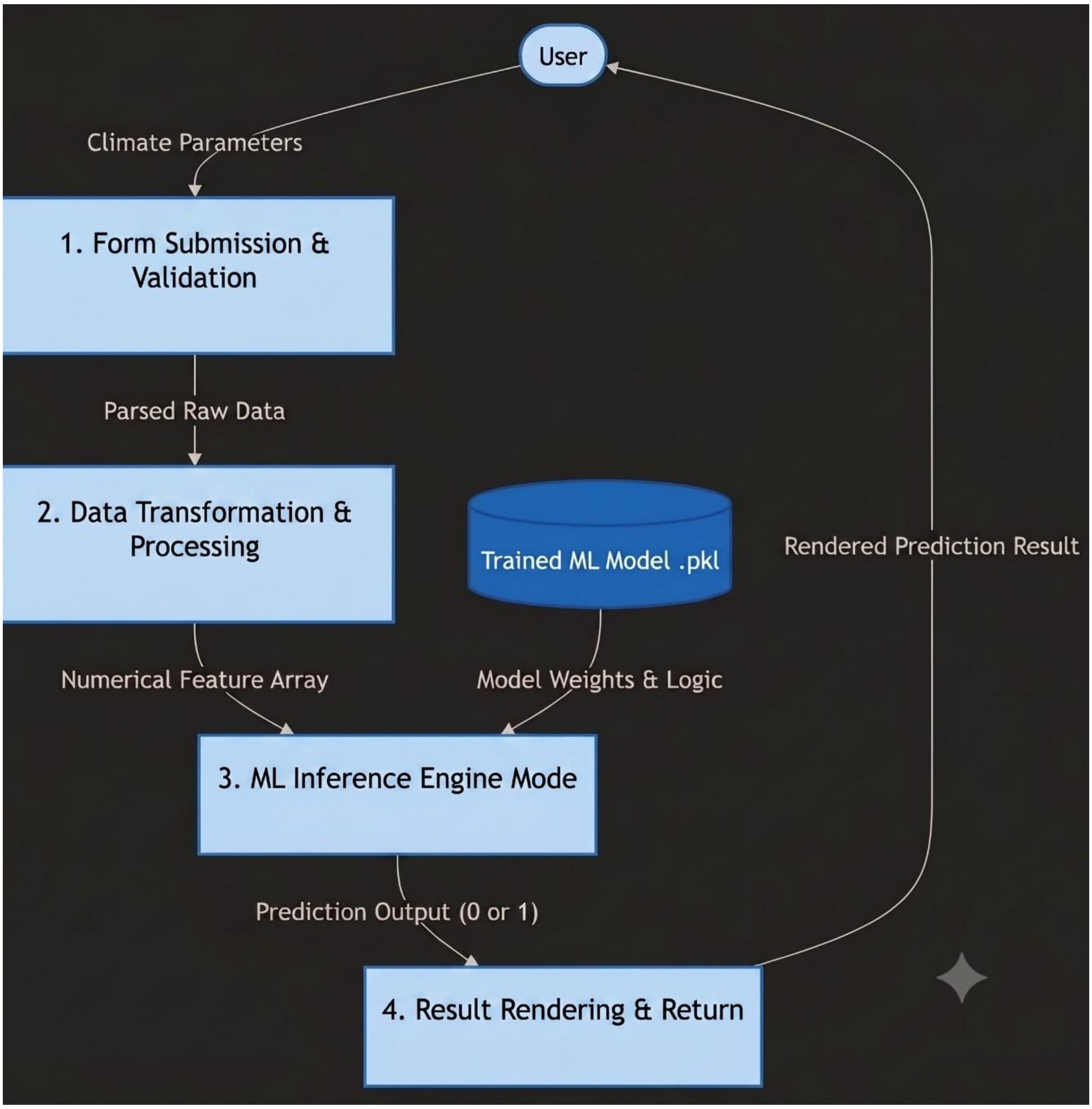
Data FlowDiagram

Date	15 March 2026
Team ID	
Project Name	Rising water
Maximum Marks	2 Marks

Data Flow Diagram



1. External Entities (Ovals)

Entity	Description
User	The person interacting with the Flask web application through their browser to enter data and receive predictions.

2. Processes (Rectangles)

Process	Description
1. Form Submission & Validation	The Flask backend (specifically the /predict route) receives the POST request from the user and extracts the variables from the form.
2. Data Transformation	The backend script structures the incoming form variables into a format the model understands (like a NumPy array or Pandas DataFrame).
3. ML Inference Engine	The core logic where the application calls .predict() on the incoming data array to determine if a flood will occur.
4. Result Rendering	The application takes the raw prediction (e.g., 0 or 1) and formats it into a user-friendly response (like rendering result.html).

3. Data Stores (Solid Fill Rectangles)

Data Store	Description
Trained ML Model (.pkl / .joblib)	The saved machine learning model file on the server's disk that holds the learned rules and algorithms. It acts as a static data store read by the Inference Engine.

4. Data Flows (Labeled Arrows)

Source	Data Passed (Label)	Destination
User	Climate Parameters	**Form Submission**
Form Submission	Parsed Raw Data	**Data Transformation**
Data Transformation	Numerical Feature Array	**ML Inference Engine**
Trained ML Model	Model Weights & Logic	**ML Inference Engine**
ML Inference Engine	Prediction Output (0 or 1)	**Result Rendering**
Result Rendering	Rendered Prediction Result	**User**